INORGANIC CUMULATIVE EXAMINATION

February 17, 1990

Inorganic Reasonings. Give explanations for eight of the following observations.

1. It requires more energy (ionization energy, ionization potential) to remove one electron from a neutral N atom than from a C atom, but it requires less energy for an O atom than a N atom.

2. The effective radius of the Fe(III) ion is larger in high spin complexes than it is in low spin complexes.

3. The 3rd ionization energy for Si is 3232 kJ/mol, the 4th is 4355 kJ/mol, and the 5th is 16091 kJ/mol.

4. The melting point of HCl is about -120, and for HF is about -80.

5. The dipole moment for SiF_4 is 0.0 but it is non-zero for SF_4.

6. BCl_3 is a better Lewis acid than BF_3.

7. H_2SeO_3 is a weaker acid than H_2SeO_4.

8. It is difficult to store aqueous solutions of the ClO^- ion.

9. Even though the second ionization energy for Ca is greater than the first, compounds of Ca^{+1} are essentially unknown.

10. All of the Lanthanide elements have a common oxidation state of +3, in spite of their differing electronic configurations.

11. In a mixture of Cr(III), NH_3, and en ($=NH_2CH_2CH_2NH_2$) all of the Cr(III) will form $Cr(en)_3^{+3}$ rather than $Cr(NH_3)_6^{+3}$ even though the bond energies for the Cr - N bonds are about the same for the two complexes.

12. More than one structure is possible for a complex of formula $M_2(CO)_8$ (M=metal).

13. Boron nitride, BN, exists in two forms: one form is an extremely hard solid, the other is a slippery solid which can be used as a lubricant.

14. White phosphorus is so reactive it must be stored under water, while red phosphorus can be stored without any special protection.

15. The chemistry of Zr is almost identical to that of Hf.

16. Fe(IV) is a fairly rare oxidation state of iron, yet FeS_2 is a common mineral.

17. There are two isomers of $PtCl_2Br_2^{-2}$ but only one of $ZnCl_2Br_2^{-2}$.

18. CO_2 has a melting point of -56 but SiO_2 has a melting point of 1713.